AI-Powered Credit Risk Intelligence Platform

github.com/Alisha-2004/credit-risk-intelligence-platform

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Problem Statement

How can financial institutions accurately predict loan default risk, provide transparent credit decisions, and enable efficient business analytics from large-scale customer data?

Objectives

- Predict the probability of customer loan default
- Generate risk scores and risk categories
- Provide explainable AI insights using SHAP
- Support business decision-making through recommendation rules
- Enable natural language data analytics using an AI-powered chatbot
- Deliver an interactive and user-friendly dashboard
- Support reproducible deployment using Docker

Proposed Solution

An AI-powered platform integrating:

✓ Credit Risk Prediction

✓ Explainable AI

✓ Risk Recommendation Engine

✓ Natural Language Analytics Assistant

Dataset

Home Credit Default Risk Dataset

Selected File: application_train.csv

Source: **Kaggle – Home Credit Default Risk**

Dataset: Home Credit Default Risk Dataset

	Metric	Value
0	Rows	307511.00
1	Columns	122.00
2	Numerical Features	106.00
3	Categorical Features	16.00
4	Default Rate (%)	8.07

Exploratory Data Analysis

Class Imbalance

- The dataset is highly imbalanced, with non-default applicants significantly outnumbering defaulters.
- This motivated the use of **SMOTE** during model training.

Credit Distribution

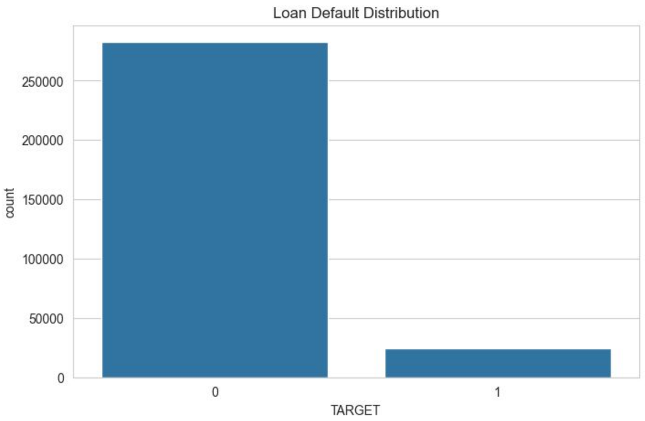
- Most applicants requested moderate loan amounts.
- A small number of customers applied for very high credit values.

Customer Demographics

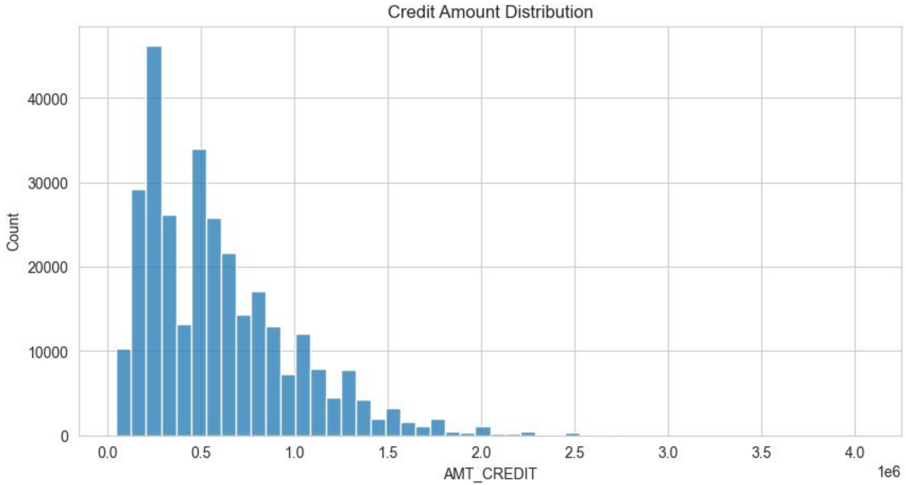
- Female applicants constitute the majority of loan applications.
- Laborers, Sales Staff, and Core Staff represent the largest occupational groups.

Business Insight

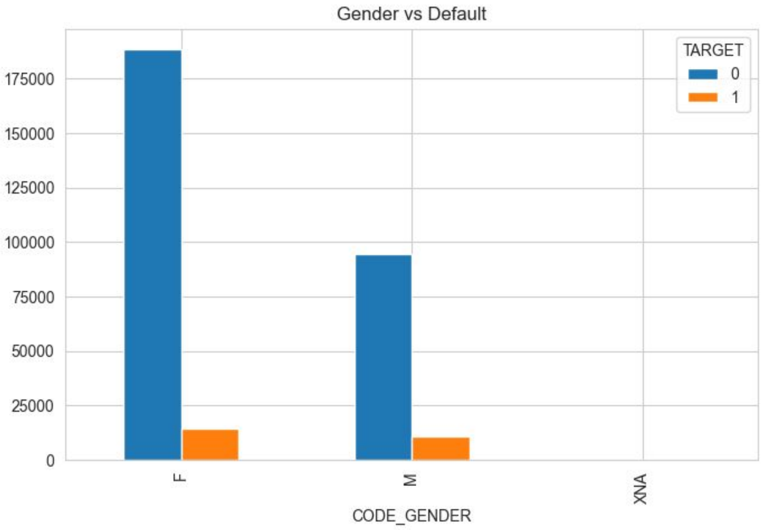
- Applicant demographics and financial attributes provide valuable signals for assessing credit risk and predicting defaults.



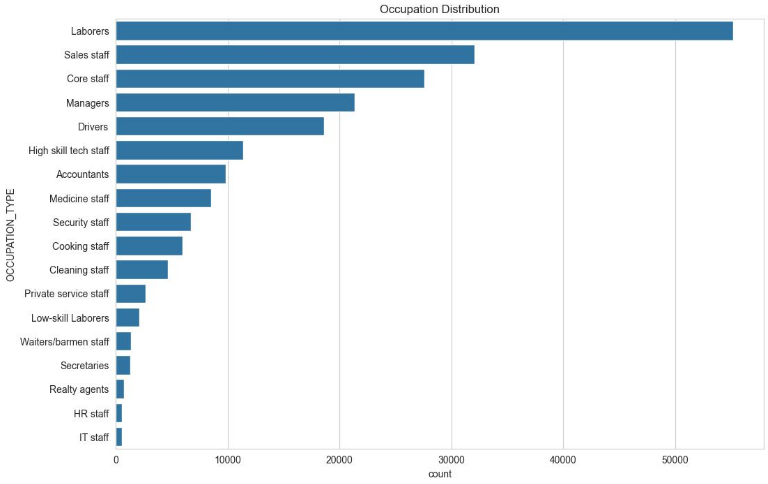
Loan Default Distribution
250000
200000
150000
100000
50000
0
count
0
1
TARGET



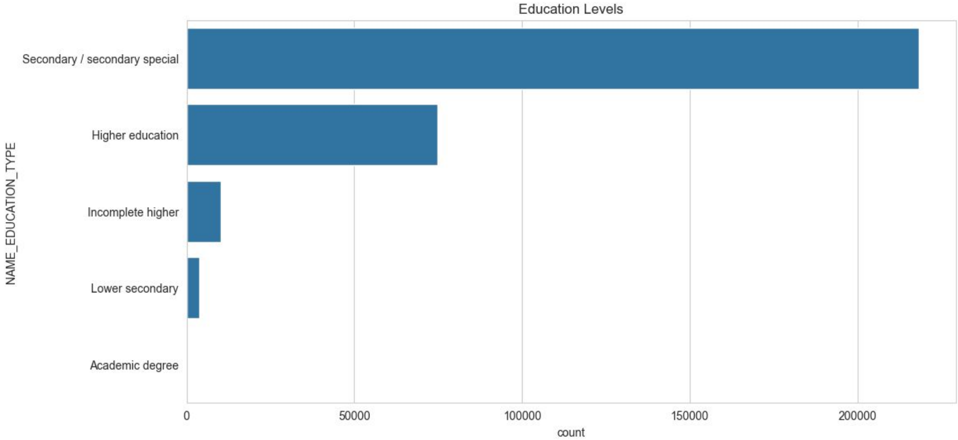
Credit Amount Distribution
40000
30000
20000
10000
0
Count
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
1e6
AMT_CREDIT



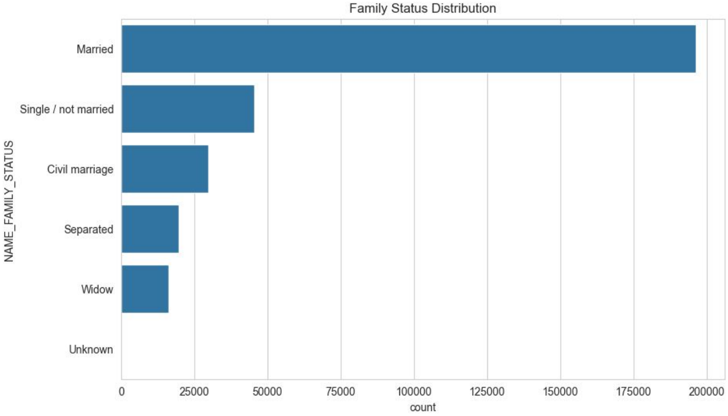
Gender vs Default
TARGET
0
1
175000
150000
125000
100000
75000
50000
25000
0
F
M
XNA
CODE_GENDER



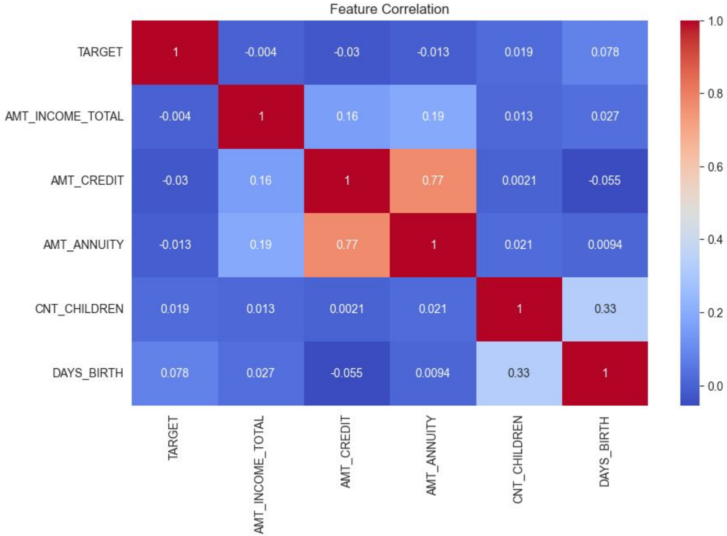
Occupation Distribution
Laborers
Sales staff
Core staff
Managers
Drivers
High skill tech staff
Accountants
Medicine staff
Security staff
Cooking staff
Cleaning staff
Private service staff
Low-skill Laborers
Waiters/barmen staff
Secretaries
Realty agents
HR staff
IT staff
OCCUPATION_TYPE
0
10000
20000
30000
40000
50000
count



Education Levels
NAME_EDUCATION_TYPE
Secondary / secondary special
Higher education
Incomplete higher
Lower secondary
Academic degree
0
50000
100000
150000
200000
count



Family Status Distribution
NAME_FAMILY_STATUS
Married
Single / not married
Civil marriage
Separated
Widow
Unknown
0
25000
50000
75000
100000
125000
150000
175000
200000
count



Feature Correlation
TARGET
AMT_INCOME_TOTAL
AMT_CREDIT
AMT_ANNUITY
CNT_CHILDREN
DAYS_BIRTH
1 -0.004 -0.03 -0.013 0.019 0.078
-0.004 1 0.16 0.19 0.013 0.027
-0.03 0.16 1 0.77 0.0021 -0.055
-0.013 0.19 0.77 1 0.021 0.0094
0.019 0.013 0.0021 0.021 1 0.33
0.078 0.027 -0.055 0.0094 0.33 1
1.0
0.8
0.6
0.4
0.2
0.0

Data Preprocessing & Feature Engineering

Data Preprocessing

- Handled missing values and data inconsistencies
- Processed numerical and categorical features
- Encoded categorical variables for machine learning
- Performed train-test split for model evaluation

Feature Engineering

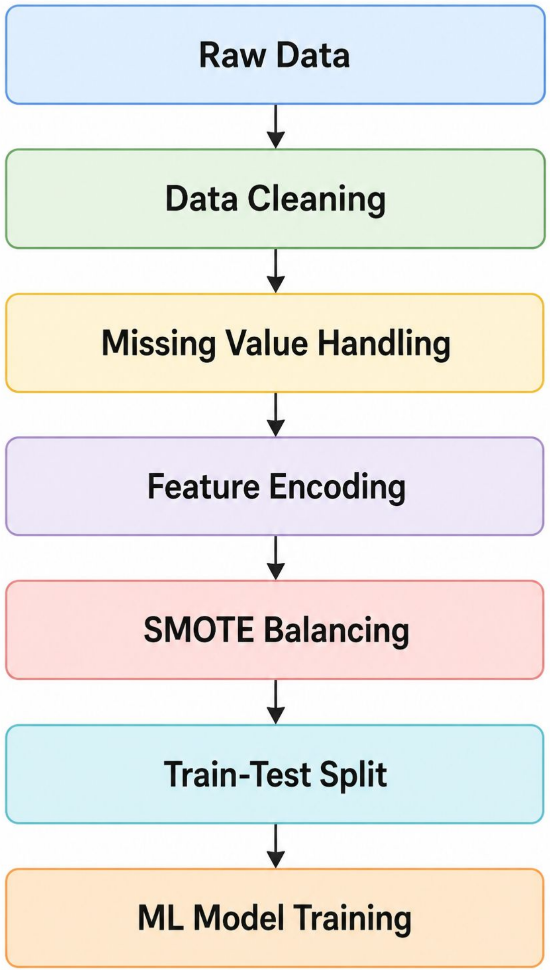
- Selected relevant financial, demographic, and credit-related attributes
- Extracted meaningful features from applicant information
- Prepared features for risk prediction and explainability

Class Imbalance Handling

- Identified severe imbalance between default and non-default classes
- Applied **SMOTE (Synthetic Minority Oversampling Technique)**
- Improved model learning on minority (default) cases

Final Dataset Preparation

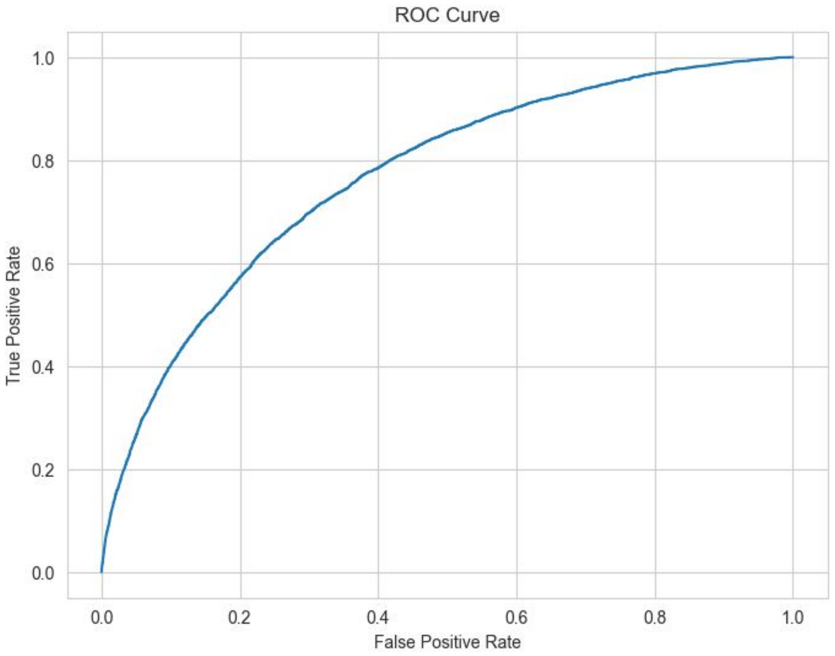
- Cleaned and transformed dataset into a machine-learning-ready format
- Generated balanced training data for robust model development

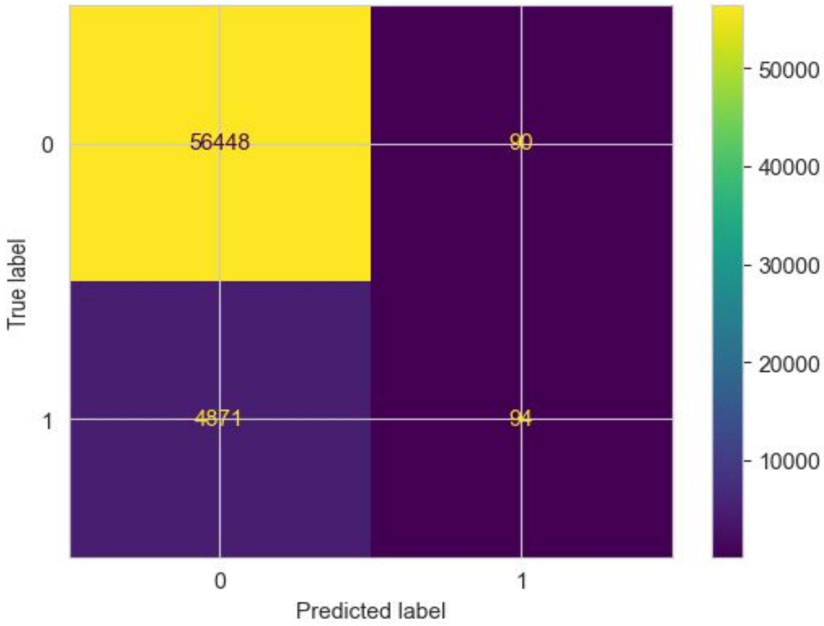

Raw Data
Data Cleaning
Missing Value Handling
Feature Encoding
SMOTE Balancing
Train-Test Split
ML Model Training

Machine Learning Model Development

- **Implemented a supervised binary classification framework** to predict loan default risk using the TARGET variable.
- **Applied SMOTE (Synthetic Minority Oversampling Technique)** to address severe class imbalance and improve minority class prediction performance.
- **Evaluated multiple machine learning algorithms** including Logistic Regression, Random Forest, and LightGBM using ROC-AUC as the primary evaluation metric.
- **Performed feature preprocessing and encoding** to transform categorical and numerical attributes into a machine-learning-ready feature space.
- **Selected LightGBM as the final model** achieving a ROC-AUC score of **0.767**, demonstrating superior predictive capability and efficient handling of large-scale tabular data.

Model	ROC-AUC
Logistic Regression	0.614
Random Forest	0.726
LightGBM	0.767





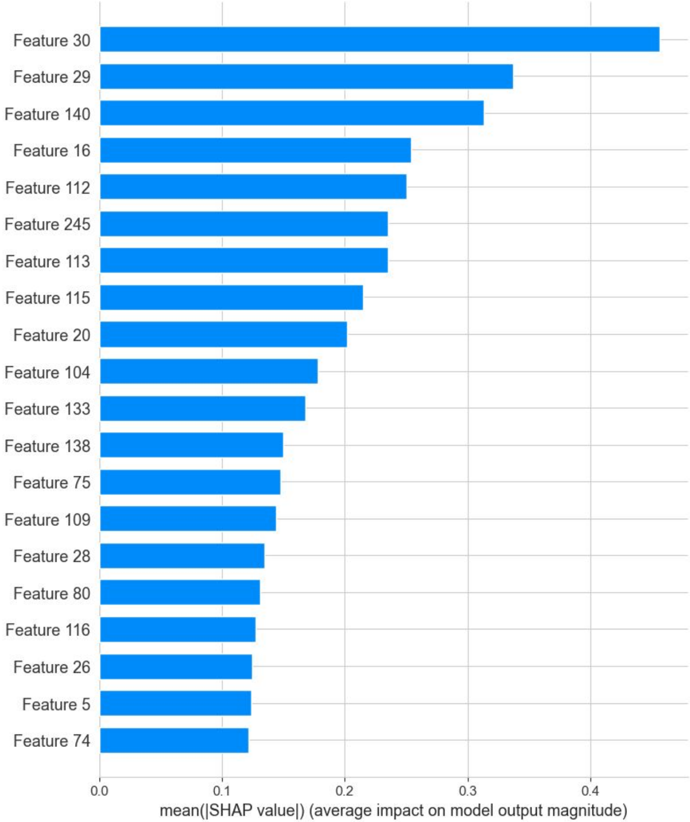
Risk Scoring & Decision Engine

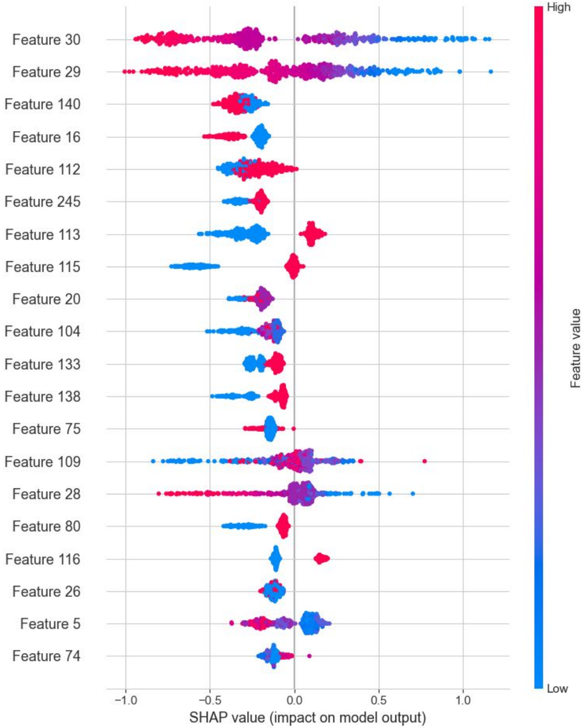
- **Converted model-predicted default probabilities into a standardized Risk Score ranging from 0 to 100.**
- **Designed a rule-based risk classification framework** to categorize applicants into Low, Medium, and High Risk groups.
- **Integrated machine learning outputs with business decision rules** to support credit approval workflows.
- **Generated interpretable lending recommendations** based on risk category thresholds and predicted default likelihood.
- **Provided a transparent decision-support mechanism** that bridges predictive analytics and banking policy requirements.

🟢 Low Risk (0–30) 🟡 Medium Risk (30–70) 🔴 High Risk (70–100)

Explainable AI

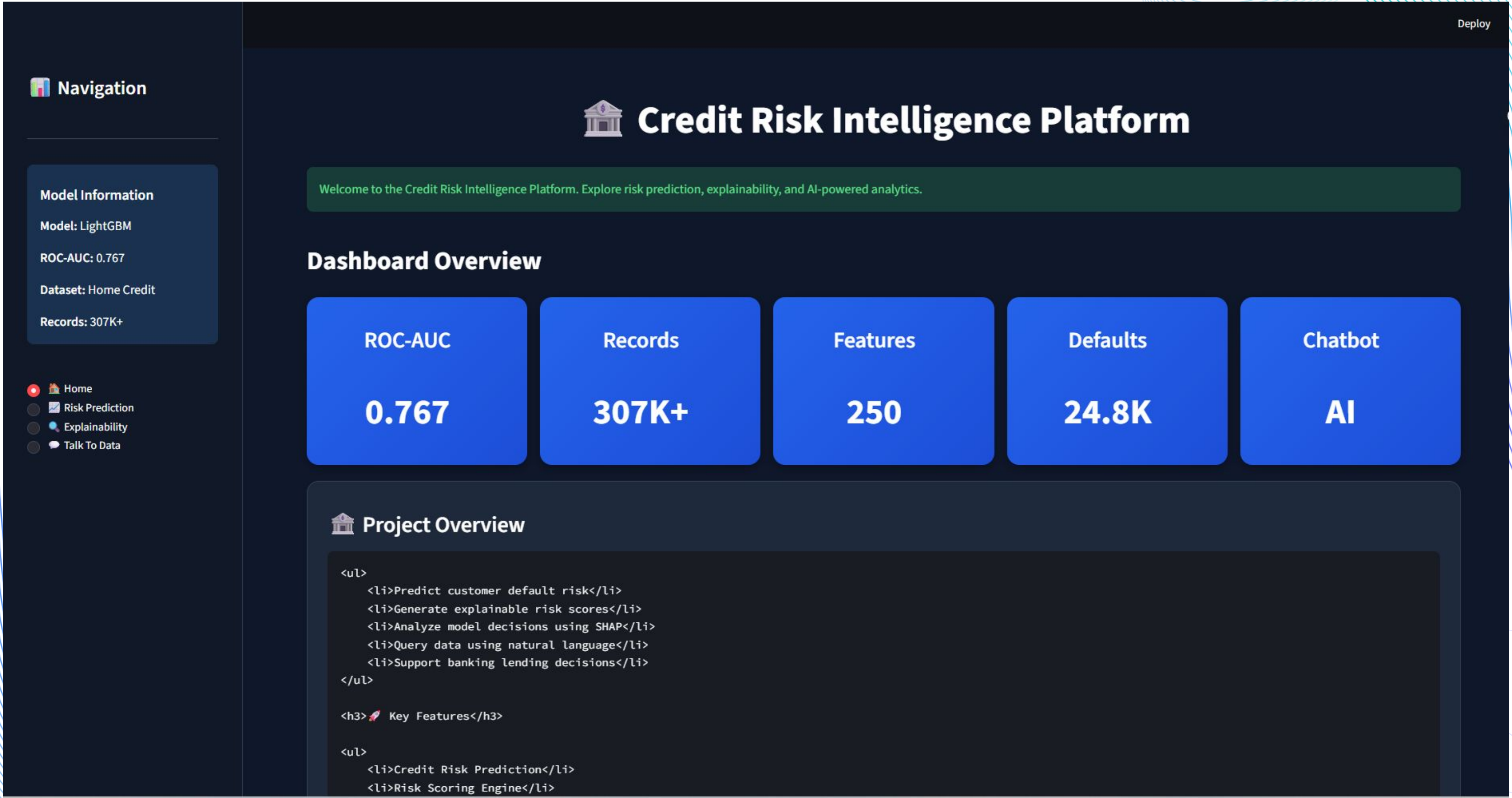
- **Applied SHAP (SHapley Additive Explanations)** to interpret predictions generated by the LightGBM model.
- **Identified the most influential features** contributing to credit risk classification and default prediction.
- **Provided global feature importance analysis** to understand overall model behavior across the dataset.
- **Enabled transparent and explainable decision-making** by quantifying each feature's impact on predictions.
- **Improved model trustworthiness and regulatory compliance** by making AI-driven credit decisions interpretable.


Feature 30
Feature 29
Feature 140
Feature 16
Feature 112
Feature 245
Feature 113
Feature 115
Feature 20
Feature 104
Feature 133
Feature 138
Feature 75
Feature 109
Feature 28
Feature 80
Feature 116
Feature 26
Feature 5
Feature 74
0.0
0.1
0.2
0.3
0.4
mean(|SHAP value|) (average impact on model output magnitude)


Feature 30
Feature 29
Feature 140
Feature 16
Feature 112
Feature 245
Feature 113
Feature 115
Feature 20
Feature 104
Feature 133
Feature 138
Feature 75
Feature 109
Feature 28
Feature 80
Feature 116
Feature 26
Feature 5
Feature 74
−1.0
−0.5
0.0
0.5
1.0
SHAP value (impact on model output)
High
Low
Feature value

AI Banking Assistant

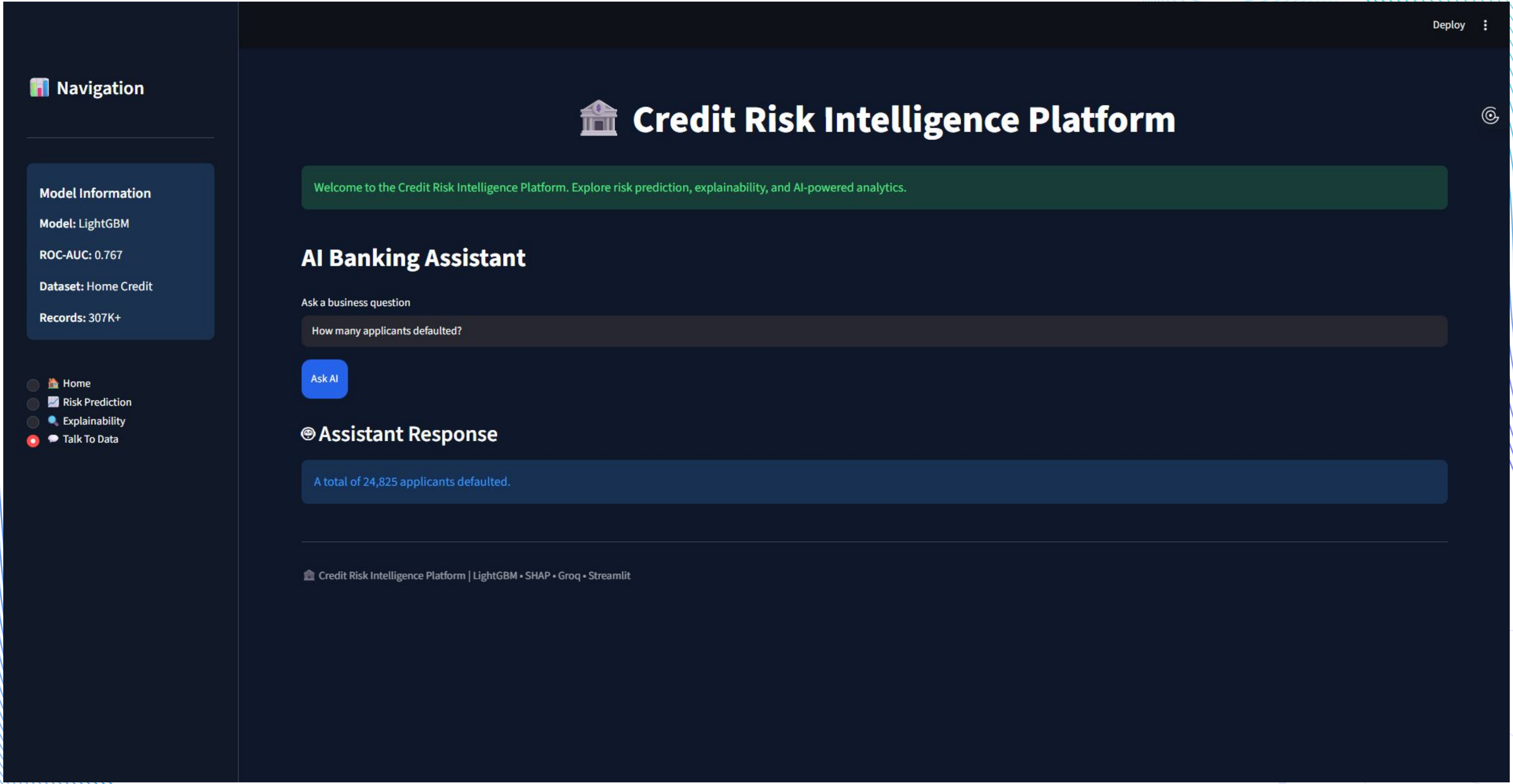
- **Developed a Natural Language to SQL (NL→SQL) pipeline** enabling users to query credit data using plain English.
- **Integrated Groq Llama 3.3** to translate user questions into executable SQL queries.
- **Connected the AI assistant with a SQLite database** containing processed credit risk data.
- **Generated business-readable insights** by converting query results into natural language responses.
- **Enabled self-service analytics** allowing non-technical users to explore credit data without SQL knowledge.


Deploy
Navigation
Model Information
Model: LightGBM
ROC-AUC: 0.767
Dataset: Home Credit
Records: 307K+
Home
Risk Prediction
Explainability
Talk To Data
Credit Risk Intelligence Platform
Welcome to the Credit Risk Intelligence Platform. Explore risk prediction, explainability, and AI-powered analytics.
Dashboard Overview
ROC-AUC
0.767
Records
307K+
Features
250
Defaults
24.8K
Chatbot
AI
Project Overview

Predict customer default risk
Generate explainable risk scores
Analyze model decisions using SHAP
Query data using natural language
Support banking lending decisions

<h3>🚀 Key Features</h3>

Credit Risk Prediction
Risk Scoring Engine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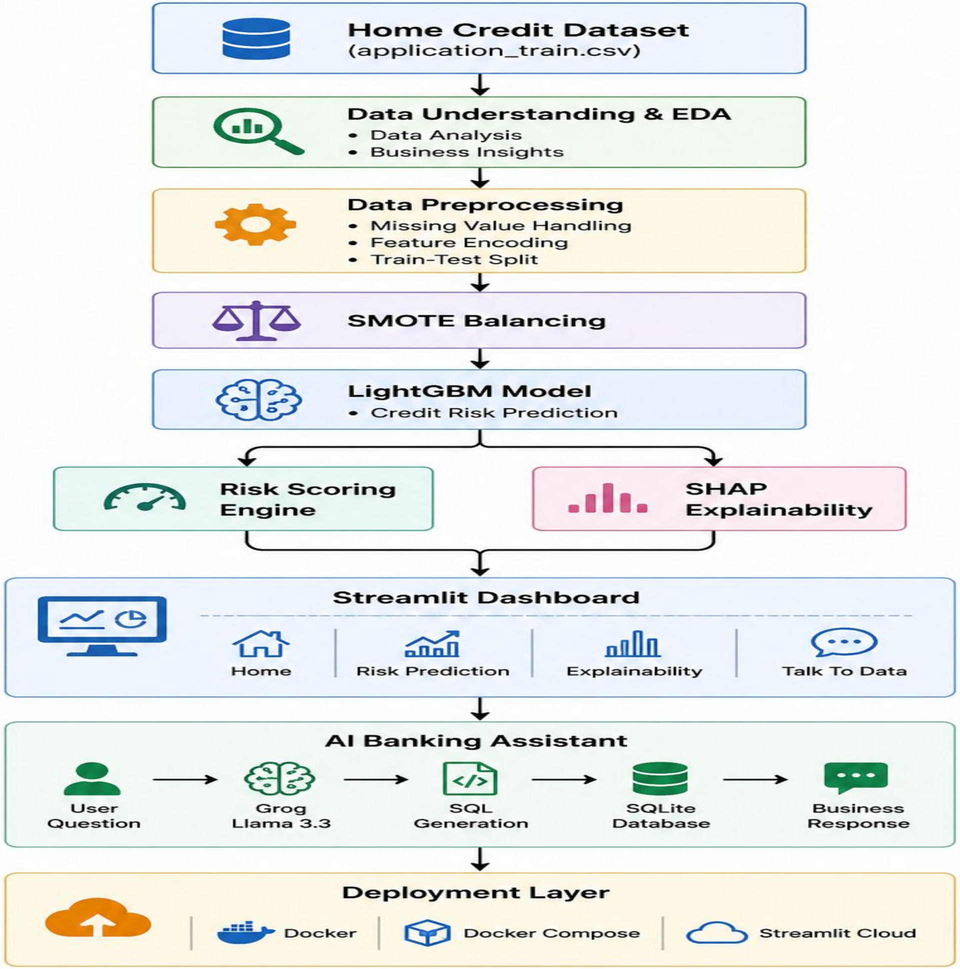
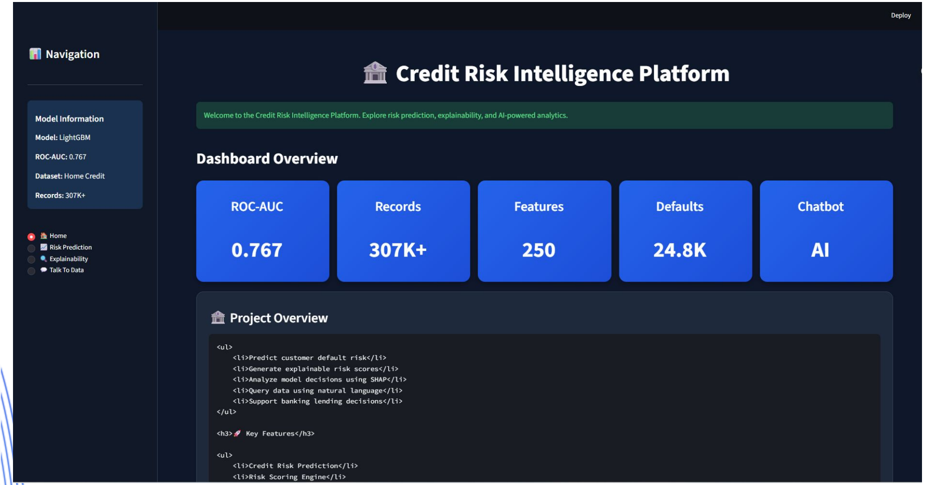
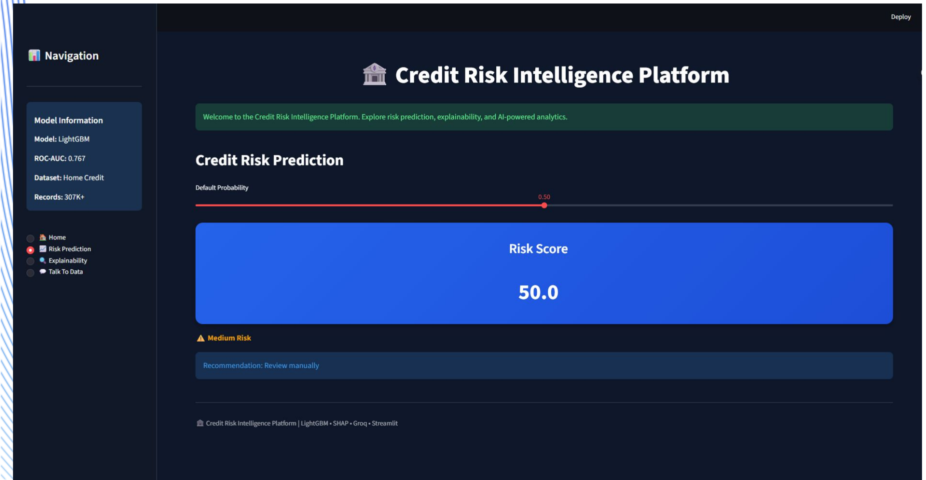
Welcome to the Credit Risk Intelligence Platform. Explore risk prediction, explainability, and AI-powered analytics.
AI Banking Assistant
Ask a business question
How many applicants defaulted?
Ask AI
Assistant Response
A total of 24,825 applicants defaulted.
Credit Risk Intelligence Platform | LightGBM • SHAP • Groq • Streamlit

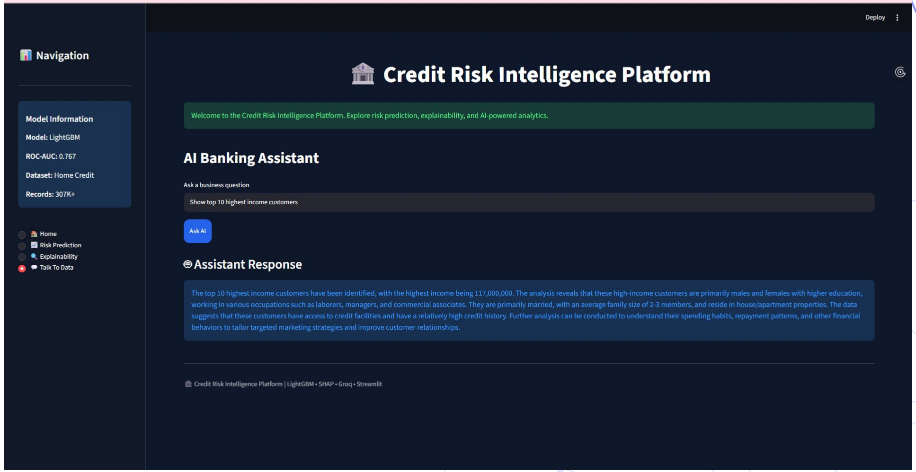
System Architecture



Dashboard Demonstration







Deployment & Engineering

- Developed a modular and scalable project architecture.
- Integrated machine learning, explainability, and NL-to-SQL components into a unified application.
- Implemented version control using **Git and GitHub**.
- Containerized the application using **Docker and Docker Compose**.
- Deployed the dashboard using **Streamlit Community Cloud**.

GitHub Repository :-
github.com/Alisha-2004/credit-risk-intelligence-platform

Alisha-2004 / credit-risk-intelligence-platform

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credit-risk-intelligence-platform

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main | 1 Branch | 0 Tags | Go to file T | Add file | Code

Alisha-2004 Final project updates and documentation improvements — 9751208 · 2 hours ago — 8 Commits

data	Final dashboard, chatbot, and documentation updates	yesterday
documents	Final dashboard, chatbot, and documentation updates	yesterday
models	Credit Risk Intelligence Platform	yesterday
notebooks	Final dashboard, chatbot, and documentation updates	yesterday
src	Final dashboard, chatbot, and documentation updates	yesterday
.dockerignore	Fix sidebar navigation	4 hours ago
.env.example	Finalize Docker deployment and application updates	13 hours ago
.gitignore	Credit Risk Intelligence Platform	yesterday
Dockerfile	Finalize Docker deployment and application updates	13 hours ago

About

No description, website, or topics provided.

- Readme
- Activity
- 0 stars
- 0 watching
- 0 forks

Releases

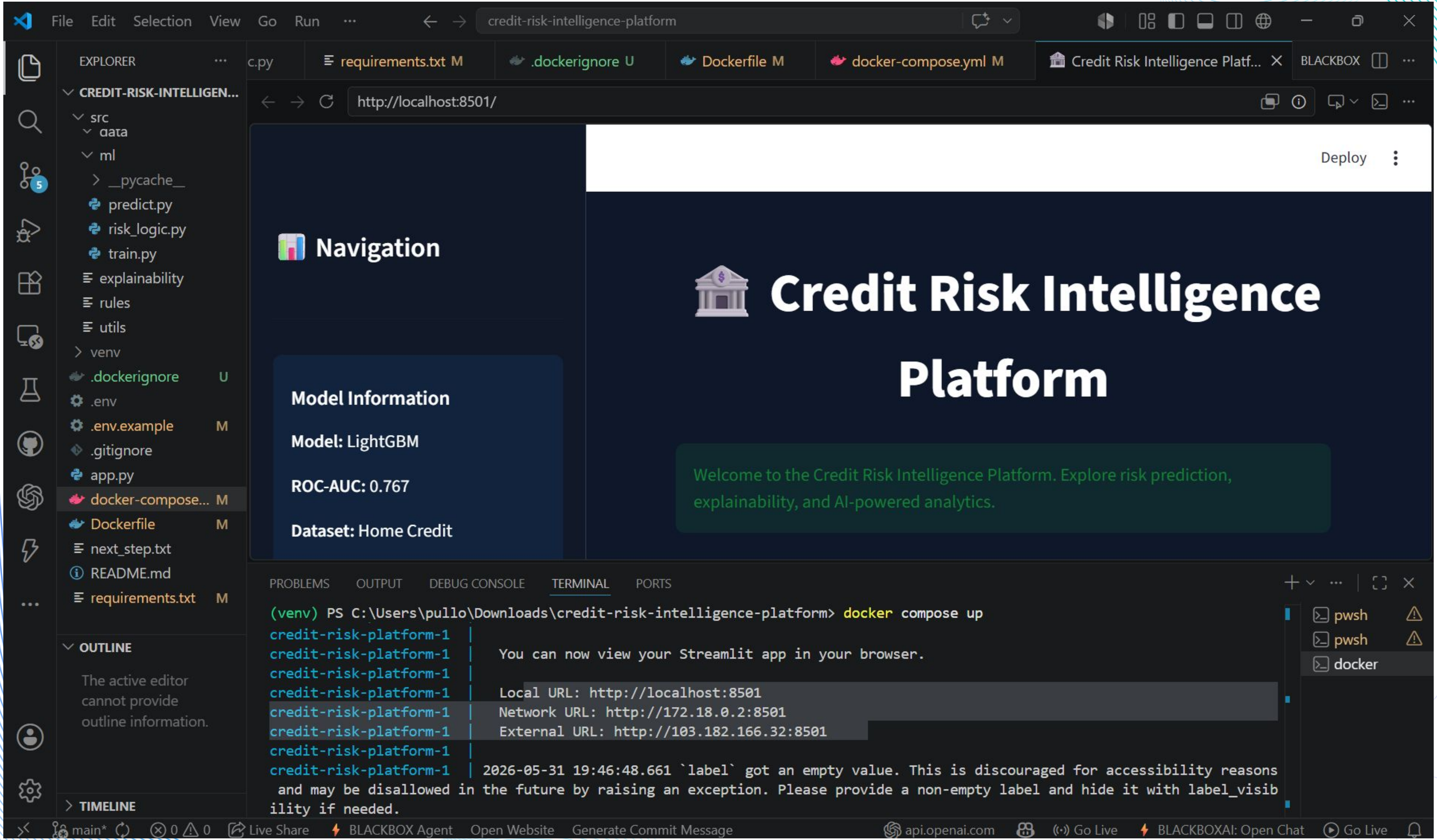
No releases published

Create a new release

Packages

No packages published

Publish your first package



File Edit Selection View Go Run ...
credit-risk-intelligence-platform
EXPLORER
CREDIT-RISK-INTELLIGEN...
src
data
ml
__pycache__
predict.py
risk_logic.py
train.py
explainability
rules
utils
venv
.dockerignore U
.env
.env.example M
.gitignore
app.py
docker-compose... M
Dockerfile M
next_step.txt
README.md
requirements.txt M
OUTLINE
The active editor cannot provide outline information.
TIMELINE
c.py
requirements.txt M
.dockerignore U
Dockerfile M
docker-compose.yml M
Credit Risk Intelligence Platf...
BLACKBOX
http://localhost:8501/
Deploy
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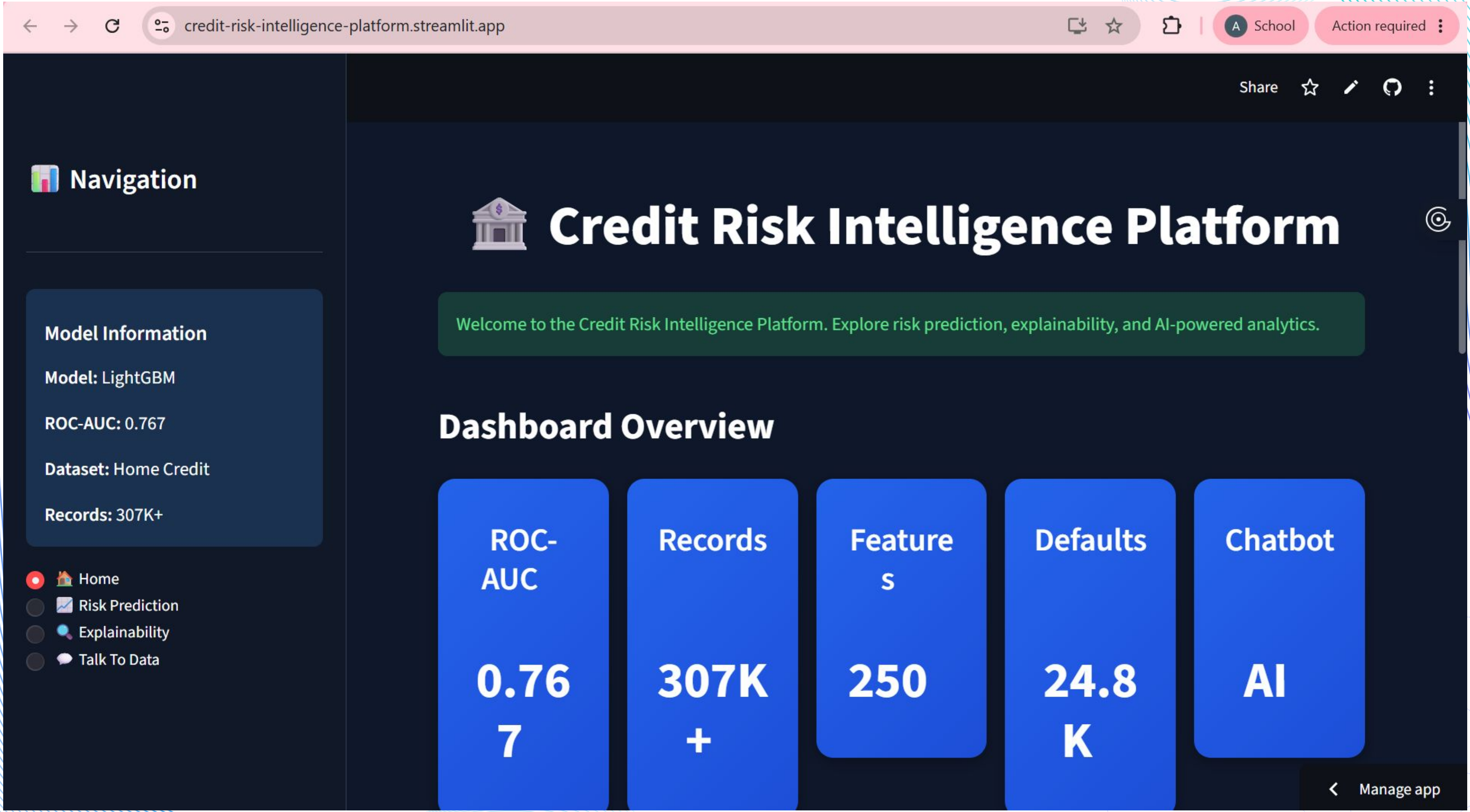
Welcome to the Credit Risk Intelligence Platform. Explore risk prediction, explainability, and AI-powered analytics.
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
(venv) PS C:\Users\pullo\Downloads\credit-risk-intelligence-platform> docker compose up
credit-risk-platform-1 |
credit-risk-platform-1 | You can now view your Streamlit app in your browser.
credit-risk-platform-1 |
credit-risk-platform-1 | Local URL: http://localhost:8501
credit-risk-platform-1 | Network URL: http://172.18.0.2:8501
credit-risk-platform-1 | External URL: http://103.182.166.32:8501
credit-risk-platform-1 |
credit-risk-platform-1 | 2026-05-31 19:46:48.661 `label` got an empty value. This is discouraged for accessibility reasons and may be disallowed in the future by raising an exception. Please provide a non-empty label and hide it with label_visibility if needed.
pwsh
pwsh
docker
main* 0 0 Live Share BLACKBOX Agent Open Website Generate Commit Message api.openai.com Go Live BLACKBOXAI: Open Chat Go Live


credit-risk-intelligence-platform.streamlit.app
School
Action required
Share
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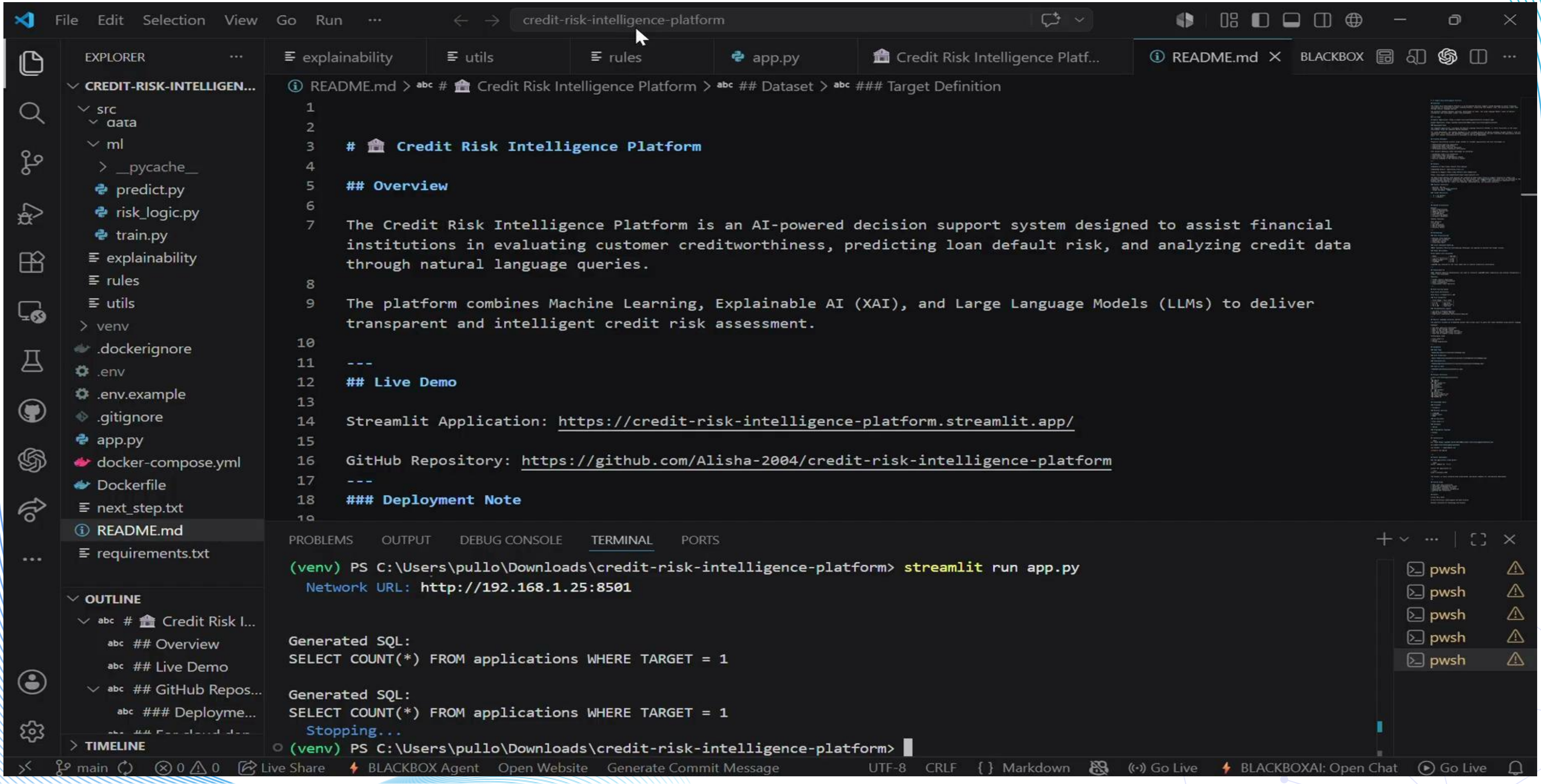
Chatbot
AI
Manage app

Live Demonstration Video

Click the video below to play



Results & Key Achievements

- Processed 307,511 loan applications
- Developed LightGBM credit risk prediction model
- Achieved ROC-AUC Score of 0.767
- Implemented SHAP-based model explainability
- Built NL-to-SQL AI Banking Assistant
- Developed end-to-end Streamlit dashboard
- Containerized deployment using Docker

THANK YOU